

Ch21: 5, 10, 24

Basic score 4` +problem 5 6`

5. (a) There is no equilibrium position for q_3 *between* the two fixed charges, because it is being pulled by one and pushed by the other (since q_1 and q_2 have different signs); in this region this means the two force arrows on q_3 are in the same direction and cannot cancel.

It should also be clear that off-axis (with the axis defined as that which passes through the two fixed charges) there are no equilibrium positions.

On the semi-infinite region of the axis that is nearest q_1 and furthest from q_2 an equilibrium position for q_3 cannot be found because $|q_2| < |q_1|$ and the magnitude of force exerted by q_1 is everywhere (in that region) stronger than that exerted by q_2 on q_3 .

Thus, we must look in the semi-infinite region of the axis which is nearest q_2 and furthest from q_1 , where the net force on q_3 has magnitude of:

$$F = k \frac{q_1 q_3}{x_3^2} + k \frac{q_2 q_3}{(x_3 - L)^2} = k q_3 \left(\frac{q_1}{x_3^2} - \frac{|q_2|}{(x_3 - L)^2} \right) \quad 2 \text{ points}$$

with $L = 10$ cm and x_3 (the x coordinate of q_3) is positive. We set this equal to zero, as required by the problem, and cancel k and q_3 . Thus, we obtain

$$\frac{q_1}{x_3^2} - \frac{|q_2|}{(x_3 - L)^2} = 0 \Rightarrow \frac{6}{x_3^2} - \frac{2}{(x_3 - L)^2} = 0 \quad 2 \text{ point}$$

$$\Rightarrow x_3 = \frac{\sqrt{3}}{\sqrt{3}-1} L = 2.37L = 23.7 \text{ cm} \quad (x = 24\text{cm, also right}) \quad 1 \text{ point}$$

for the distance between q_3 and q_1 . That is, q_3 should be placed at $x = 23.7$ cm along the x -axis.

(b) As stated above, $y = 0$. 1 point

10. If we take each angle θ in the figure to be positive, then we have $\cos\theta = R/r$ (where r is the distance between ions 3 (or 4) and electron 2, the dashed line distance shown in the figure). From symmetry, we see that there is no net force in the vertical direction on $q_2 = -e$ sitting at a distance R to the left of the coordinate origin. We note that the net x force caused by q_3 and q_4 on the y axis will have a magnitude equal to

$$2 \frac{qe}{4\pi\epsilon_0 r^2} \cos\theta = \frac{2qe \cos\theta}{4\pi\epsilon_0 (R/\cos\theta)^2} = \frac{2qe \cos^3\theta}{4\pi\epsilon_0 R^2} .$$

Consequently, to achieve a zero net force along the x axis, the above expression must equal the magnitude of the repulsive force exerted on q_2 by $q_1 = -e$. Thus,

$$\frac{2qe \cos^3 \theta}{4\pi\epsilon_0 R^2} = \frac{e^2}{4\pi\epsilon_0 R^2} \Rightarrow q = \frac{e}{2 \cos^3 \theta}.$$

The requirement that $q \leq 5e$ leads to

$$\frac{e}{2 \cos^3 \theta} \leq 5e \Rightarrow \frac{1}{(10)^{1/3}} \leq \cos \theta$$

which yields $\theta \leq 62.34^\circ$. The problem asks for “physically possible values,” and it is reasonable to suppose that only positive-integer-multiple values of e are allowed for q . If we let $q = ne$, for $n = 1 \dots 5$, then θ_n will be found by taking the inverse cosine of the cube root of $(1/2n)$.

$$\frac{e}{2 \cos^3 \theta} \leq 5e \Rightarrow \cos \theta = \frac{1}{(10)^{1/3}}, \frac{1}{(8)^{1/3}}, \frac{1}{(6)^{1/3}}, \frac{1}{(4)^{1/3}}, \frac{1}{(2)^{1/3}}$$

(a) The largest value of angle is $\theta_1 = 62.3^\circ$ (or 1.088 rad).

(b) The second largest value of angle is $\theta_2 = 60^\circ$ (or 1.047 rad).

(c) The third largest value of angle is $\theta_3 = 56.6^\circ$ (or 0.988 rad).

24. We start with spheres 1 and 2, each having charge q and experiencing a mutual repulsive force:

$$F = kq^2 / r^2.$$

First touch—when the neutral sphere 3 touches sphere 1, sphere 1’s charge decreases to $q/2$.

Second touch—then sphere 3 (now carrying charge $q/2$) is brought into contact with sphere 2; a total amount of $q/2 + q$ becomes shared equally between them. Therefore, the charge of sphere 3 and sphere 2 both are $3q/4$.

Third touch—then sphere 3 ($3q/4$) is brought into contact with sphere 1 again, a total amount of $q/2 + 3q/4$ becomes shared equally between them. Therefore, the charge of sphere 1 is $5q/8$. The repulsive force between spheres 1 and 2 is finally

$$F' = k \frac{(5q/8)(3q/4)}{r^2} = \frac{15}{32} k \frac{q^2}{r^2} = \frac{15}{32} F \Rightarrow \frac{F'}{F} = \frac{15}{32} = 0.47$$